The Delighful Dishes of Combinatorics

Problem 1

How many 6-digit numbers have strictly-decreasing digits when read from left to right?

Problem 2

How many 6-digit numbers have non-decreasing digits when read from left to right?
Hint: Consider a similar approch to stars and bars.

Problem 3

Suppose that $k, n \in \mathbb{Z}$.

(a) How many binary numbers of length at most n exist with exactly k 1's?

(b) How many binary numbers of length at most n exist with at least k 1's? (Closed form not expected here.)

Problem 4

King Justin IV is climbing pizza-decorated n stairs of his Syldavian Palace with steps of 1 or 2. How many unique combinations of steps can he take (n one step, $n-2$ one step and 1 two steps, etc)?

Problem 5

Prove that if m, n are positive integers and $1 \leq k \leq n$, then

$$\binom{n}{0}\binom{m}{k} + \binom{n}{1}\binom{m}{k-1} + \cdots + \binom{n}{k}\binom{m}{0} = \binom{m+n}{k}.$$

Hint: Consider selecting from a set of people of size $n+m$ that has n male and m female.

Problem 6

Suppose that $n \geq m$, Prove that

$$\sum_{k=m}^{n} \binom{n}{k}\binom{k}{m} = 2^{n-m}\binom{n}{m}.$$

Hint: Consider the number of subsets of sets of size n.

Problem 7

At a research station on The Skeld, Captain Red has $(n+1)$ beans, n Crewmates of distinct colors and one **Morphling** who shapeshifts to mimic the color of anyone it spends the night with. Captain Red selects 4 beans at random and locks them in Security overnight. The next day, he must split them into two separate pairs such that **no pair contains two beans of the same color**. Prove that the number of possible color pairings is:

$$3\binom{n+1}{4} = \binom{\binom{n}{2}}{2}.$$

Problem 8

Consider the set $2^{[n]}$, where $[n] := \{1, 2, \ldots, n\}$ is the index set of n elements. Prove that the minimum number of chains $\mathcal{C}_i = \{C_1, C_2, \ldots, C_m\}$, where

$$C_1 \subseteq C_2 \subseteq \cdots \subseteq C_m,$$

required to cover $2^{[n]}$ is at least $\binom{n}{\lfloor \frac{n}{2} \rfloor}$.

Hint: If $A, B \subseteq [n]$ are such that $|A| = |B|$, then we cannot have $A \subseteq B$ or $B \subseteq A$.

Problem 9

Let S be a set of n^2+1 points in $\mathbb{R}^2$. Prove that there is a sequence of $n+1$ points

$$(x_1, y_1), \ldots, (x_{n+1}, y_{n+1}) \in S \text{ such that } x_1 \leq x_2 \leq \cdots \leq x_{n+1},$$

and either

$$y_1 \geq y_2 \geq \cdots \geq y_{n+1} \text{ or } y_1 \leq y_2 \leq \cdots \leq y_{n+1}.$$

Problem 10

A 6 bead white coloured necklace is being coloured into blues, blacks, or browns. Given all beads are coloured, how many unique combinations exist?

Problem 11

An 8 digit binary sequence is constructed by choosing 0s or 1s for each digit. How many unique binary digit sequences can be created, noting that rotating (shifting last digit to the front) can create symmetries?

Problem 12

Master Chen is hosting a round-robin tournament between 4 players. In how many ways can he host the tournament, which is a collection of matches, where a match is exclusively between two players that are different from each other and each players plays each other regardless of previous matches?